VICTORY 2027

Water Resources

LECTURE-1 Geography

Sameer Sir



Topics To Be Covered

1. Sources
2. Need for conservation
3. Rain water harvesting
4.

Water Resources

- *Sources (Surface water and ground water).*
- *Need for conservation*
- *Rain water harvesting and its importance.*
- *Irrigation: Importance and methods – wells, tube well, tanks, canal, drip and sprinkler irrigation.*





i) Need for water Conservation → 5 points

ii) Methods of water Conservation → 3 points

1) Rain Water Harvesting
2) 3 R's
   - Reduce
   - Reuse
   - Recycle.
3) Reducing water pollution.

Telegram

https://t.me/icsebysameersir



Sources of Water

- **The term 'water resource' refers to any of the entire range of natural waters that occur on the earth and are of potential use to living beings.**
- Of these resources the ones which are most available for use are the waters of oceans, rivers and lakes.
- Other available water resources comprise groundwater, deep sub-surface waters and glaciers and permanent snowfields.

Sources of Water

- About 97 per cent of the earth's water supply is in the ocean which is unfit for human consumption and other uses due to high salt content. → Salinity
- Of the remaining 3 per cent, 2.3 per cent is locked in the polar ice caps.
- The balance 0.7 per cent is available as freshwater but 0.66 per cent is groundwater and the rest 0.03 per cent is available to us as freshwater in rivers, lakes and streams.

Sources of Water

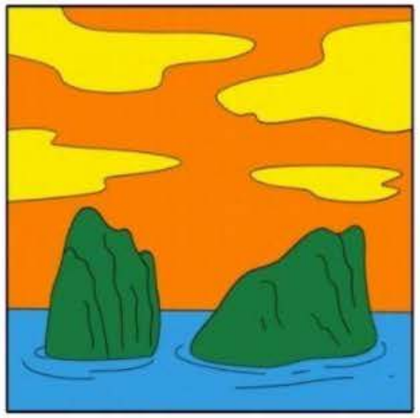
बारिश

(a) **Surface Water:** Water on land is the result of precipitation or seepage from underground which forms streams, lakes, rivers and ponds flowing on land and finally joining the sea.

- **Surface water is any body of water found on the Earth's surface, including rivers, streams, lakes, reservoirs, ponds, wetlands, and oceans.**



Lakes or Ponds



Sea or Ocean

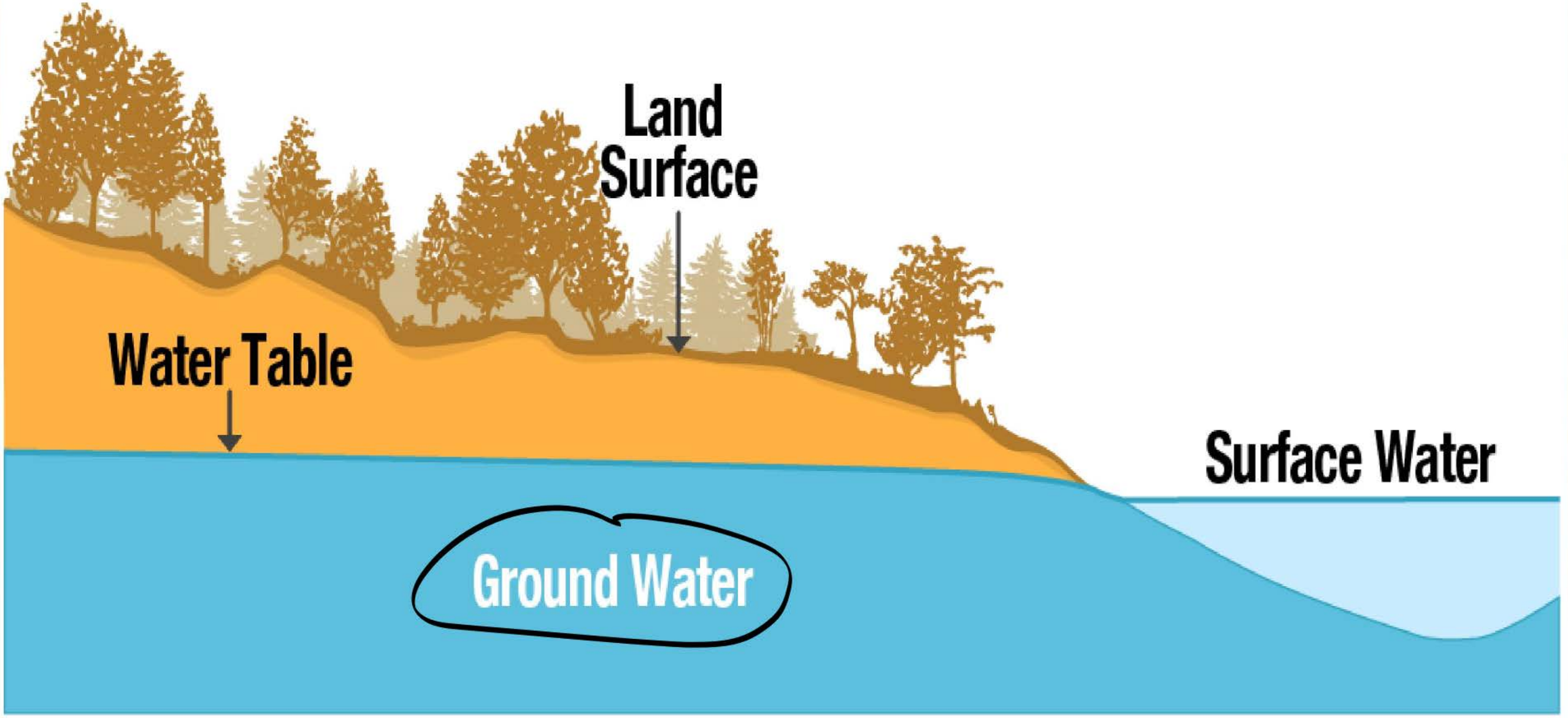
SURFACE SOURCES OF WATER

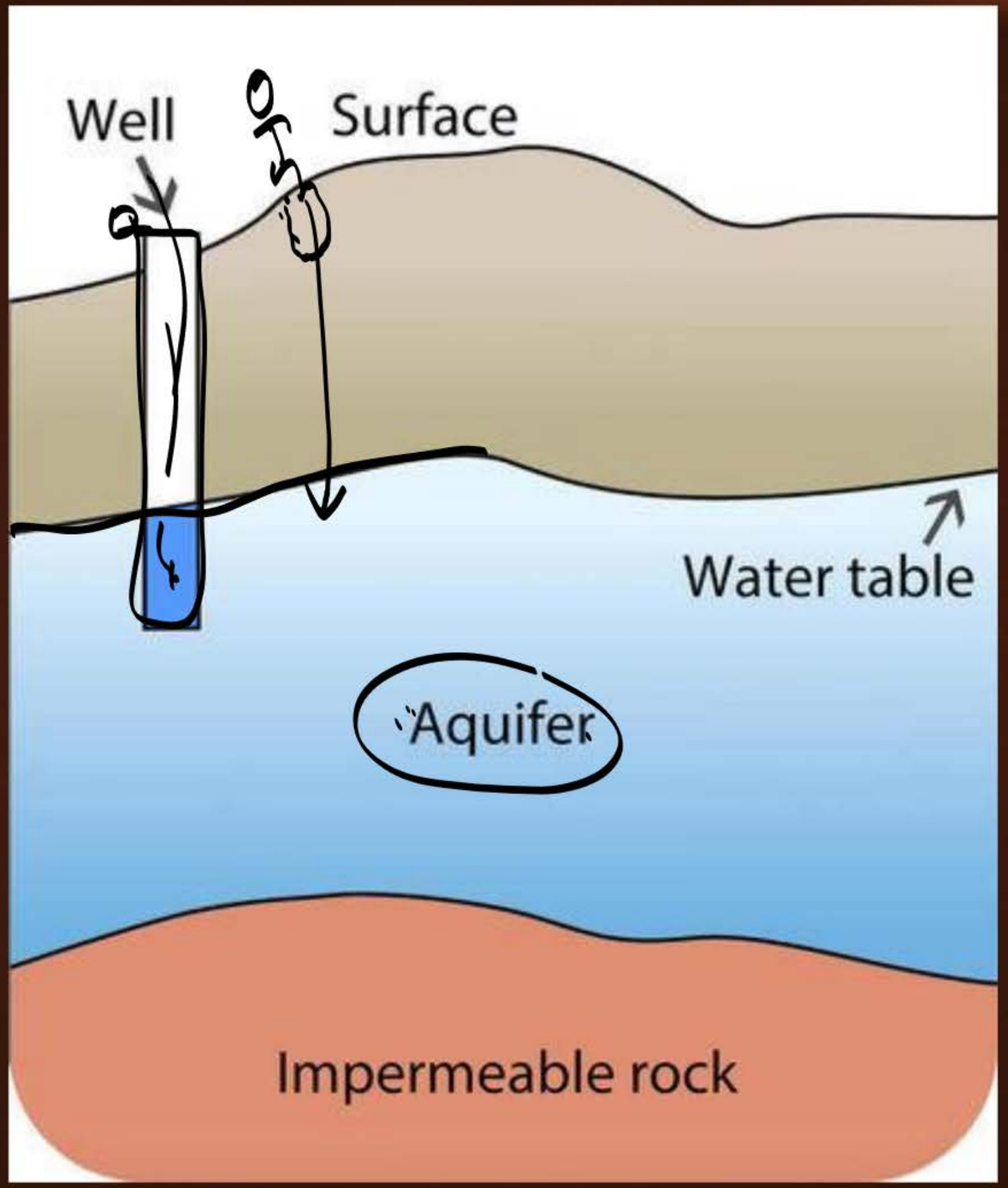


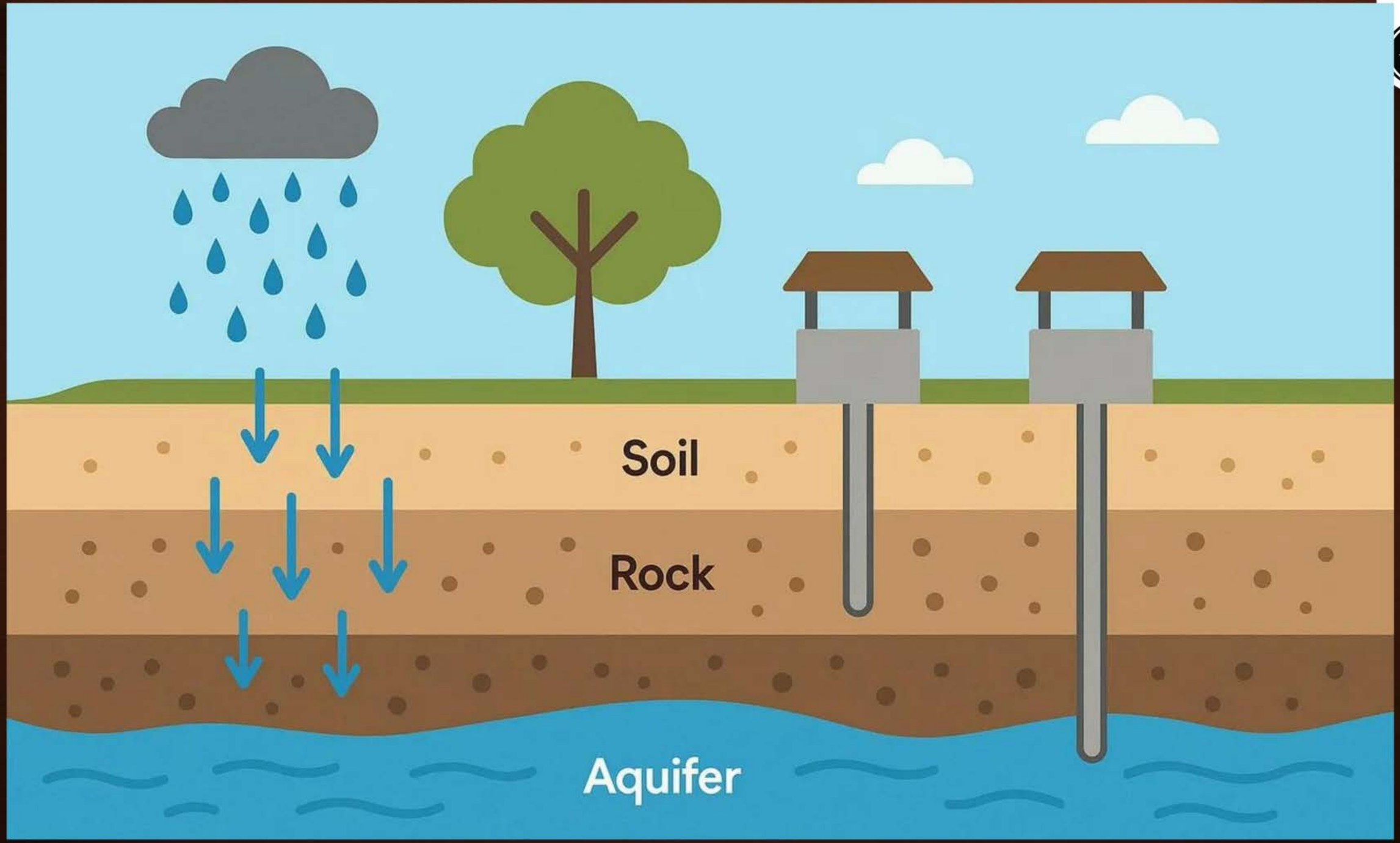
River



Reservoir


Land
Surface
Water Table
Surface Water
Ground Water


Well
Surface
Water table
Aquifer
Impermeable rock


Soil
Rock
Aquifer

Sources of Water

Aquifer → Natural + Underground Storage of water between the porous rock layer that can be utilised via tubewell or well.

(b) **Groundwater:** The water from precipitation, that is from rain or snow, flows on the surface. Some of the water evaporates and the rest sinks into the soil. The water which seeps through the cracks and crevices under the surface of the land is known as groundwater. Most of the water ultimately reaches the sea. Groundwater constitutes 0.66 per cent of usable water on the earth.

An aquifer is an underground layer of rock, sand, or gravel that holds water like a giant natural sponge and allows water to be pumped out through wells.

Sources of Water

- Nature has endowed India with plentiful water resources.
- These resources account for four per cent of the world's water resources.
- Besides surface water, i.e., from rivers, lakes, ponds and tanks and groundwater resources, India has a vast coastline with indented coasts in some states.

Sources of Water

- Odisha, Kerala and West Bengal have vast surface water resources in the form of lagoons and lakes.

Brackish
Salty water
- More than fresh water
- Less than ocean water

- Though water is brackish in these water bodies, it is utilised for fishing and irrigation of certain crops like paddy, coconut, etc.

Need To Conserve Water

- Water is a natural resource which satisfies basic human need.
- Due to increase in population, irrigation and industrialisation, the demand for water has risen.
- This has led to a decline in groundwater levels in various parts of the country.

Need To Conserve Water

- According to a report by Central Ground Water Board (CGWB), in recent years, Delhi, Haryana, Chandigarh, Himachal Pradesh and Uttarakhand have recorded a sharp fall in the groundwater levels, ranging from 70 to 80 per cent.

- If this trend continues, there will be a critical shortage of water of suitable quality to meet the future needs. We need to conserve water for the following reasons:

Need To Conserve Water

We need to conserve water for the following reasons:

(i) The overexploitation of groundwater often results in the lowering of water table.

(ii) The loss of vegetation causes drought and reduction of rainfall.

(iii) Irrigation utilises more than 90per cent of the total freshwater.

Need To Conserve Water

(iv) The increase in population results in water scarcity. → कमी

(v) Our water resources like the groundwater, rivers, lakes etc. are polluted and their water can hardly be used without adequate treatment.

Conservation Practices

Rain water Harvesting → Conservation

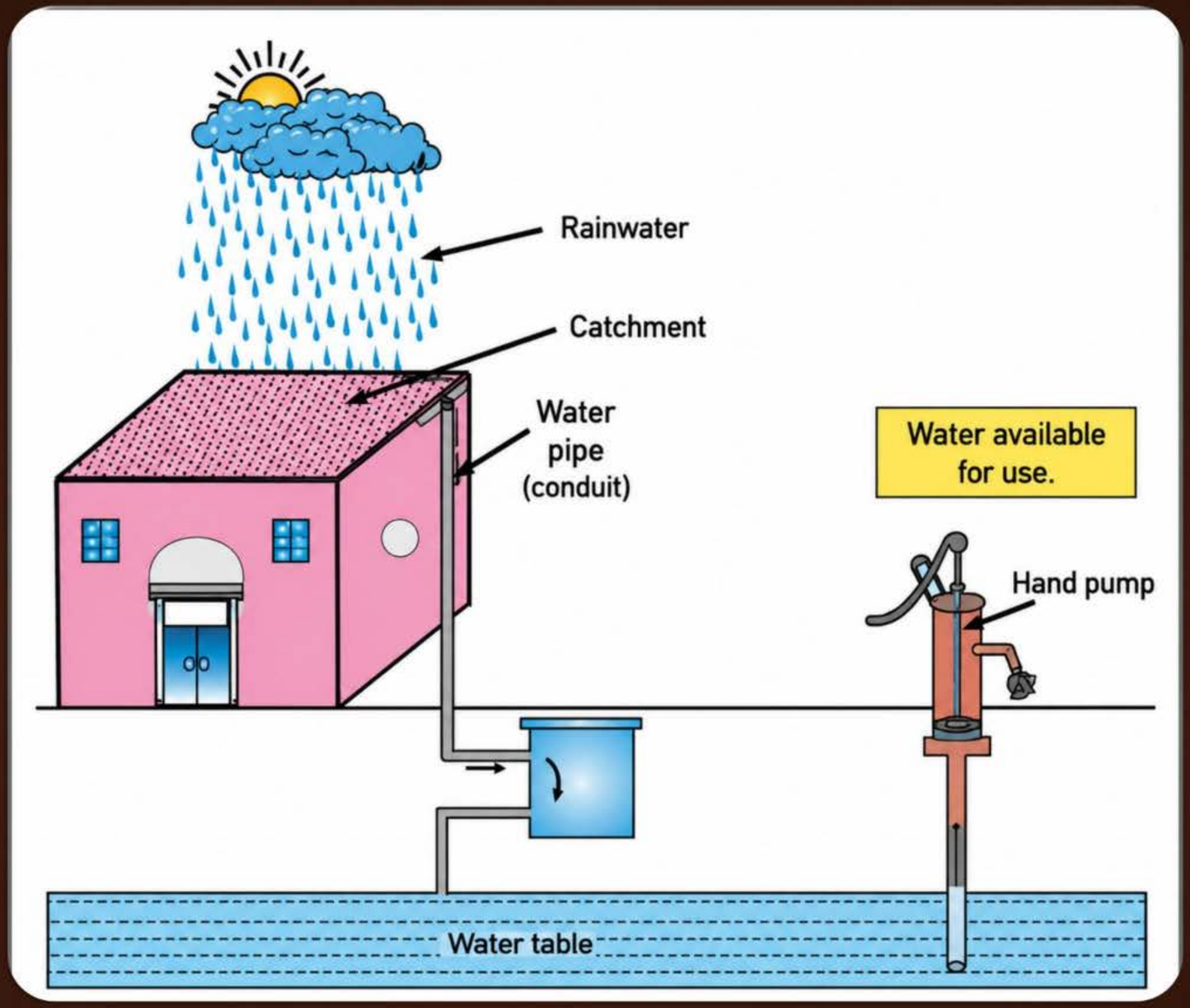
- Due to declining availability of freshwater and increasing demand, we need to conserve water and manage our water resources.
- This can be done by rainwater harvesting, adopting water saving technologies, water recycling and preventing the pollution of water.

Conservation Practices

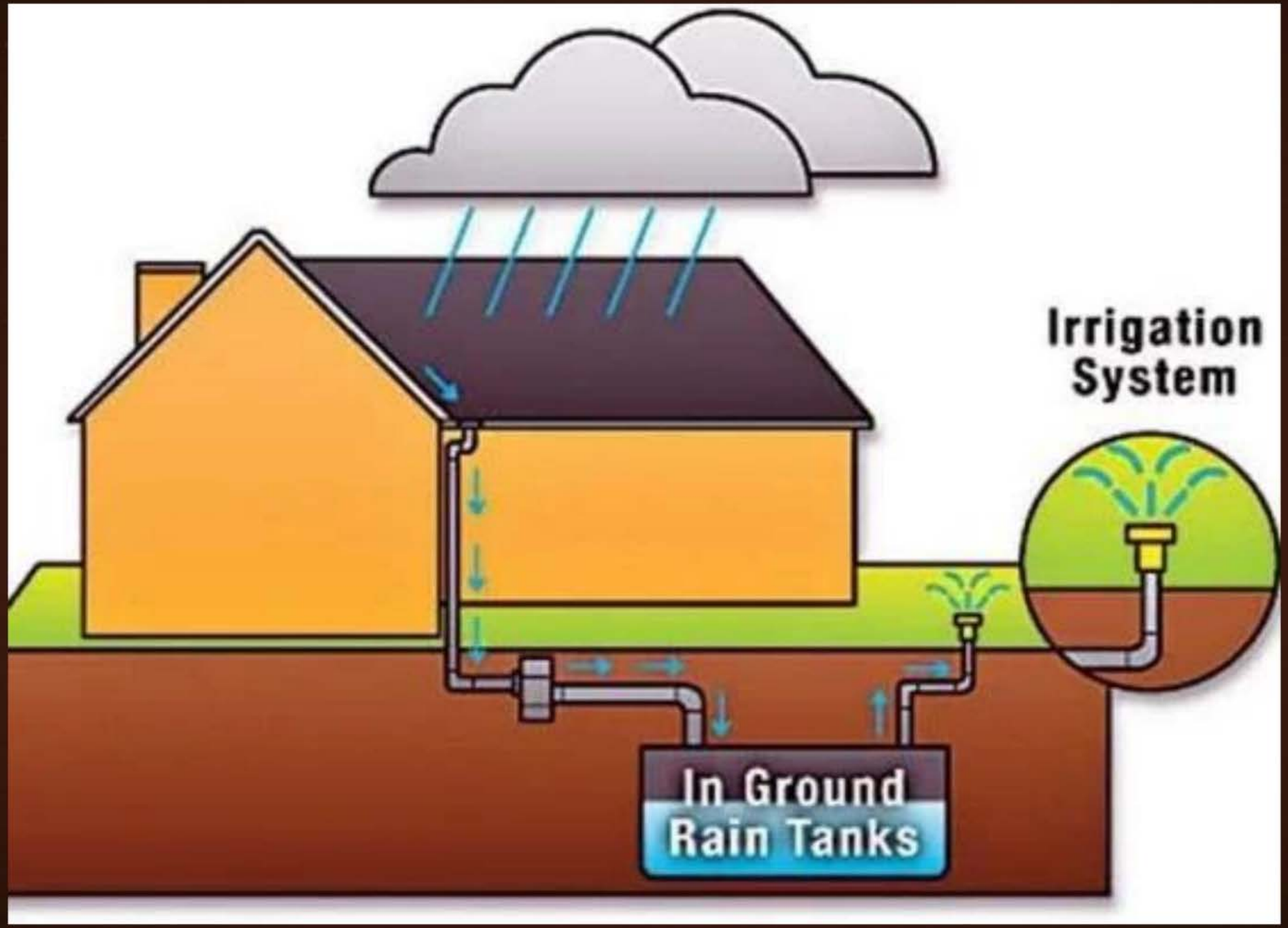
Rainwater Harvesting

- Rainwater harvesting is the collection and storage of rain from surfaces like roofs instead of letting it run off and go to waste.

(i) harvesting surface and groundwater,
(ii) prevention of losses through evaporation and seepage, and



Modern Rainwater Harvesting


Irrigation System
In Ground Rain Tanks

Objectives of Rainwater Harvesting

The main objectives of rainwater harvesting are the following:

- Meeting the ever increasing demand for water.
- Reducing the run-off which chokes drains.
- Avoiding the flooding of roads.
- Augmenting the groundwater storage and raise the water table.
- Reducing groundwater pollution.
- Improving the quality of groundwater.
- Reducing the soil erosion.
- Supplementing domestic water requirement during summer and drought.

Conservation Practices

- Rain is the primary source of water according to the hydrological cycle. → Water Cycle
- Water harvesting means understanding the value of rain and making optimum use of rainwater at the place where it falls.
- Water harvesting is the activity of direct collection of rainwater that can be stored for direct use or can be used to recharge groundwater.

Conservation Practices

- Water has been harvested in India from ancient times. Our ancestors had perfected the art of water management.
- Many water harvesting structures and water conserving mechanisms were followed in each region of the country.

Conservation Practices

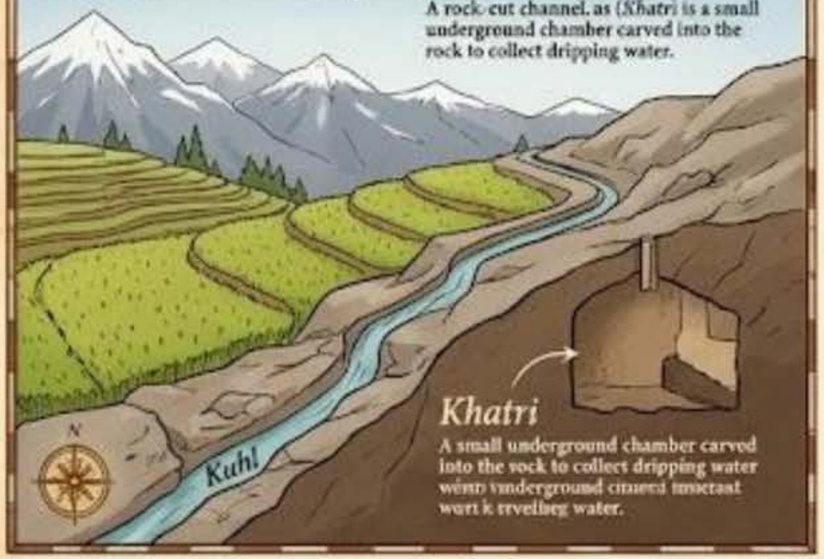
- The storing mechanisms or the water receptacle was known by different names in different areas.

- They were called
  - Khatri or Kuhl in Western Himalayas,
  - Baolis or Dighis in the Gangetic plains,
  - Johads in Central India,
  - Bhandaras or Kere in the Deccan Plateau,
  - Surangam in Western Ghats,
  - Korambu in Eastern Ghats and
  - Zing in Ladakh.

TRADITIONAL WATER HARVESTING SYSTEMS OF INDIA

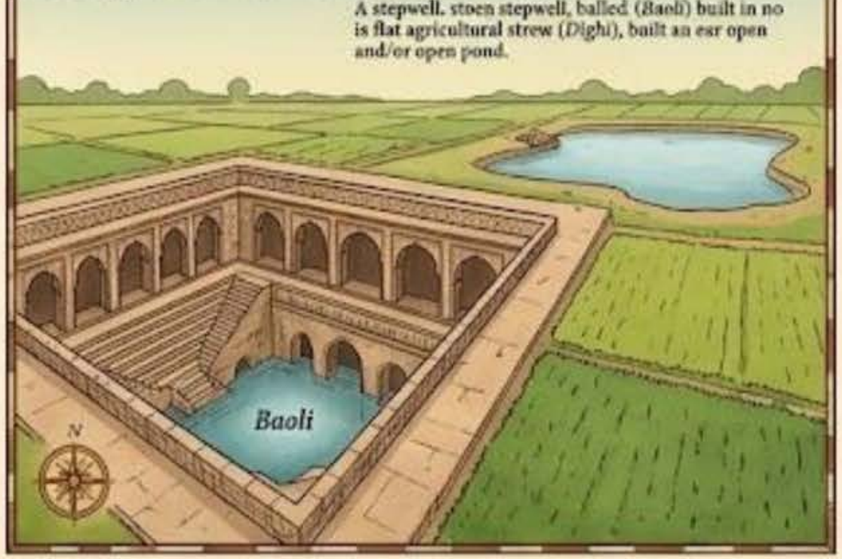
Western Himalayas: *Khatri/Kuhl*

A rock-cut channel. as (*Khatri* is a small underground chamber carved into the rock to collect dripping water.



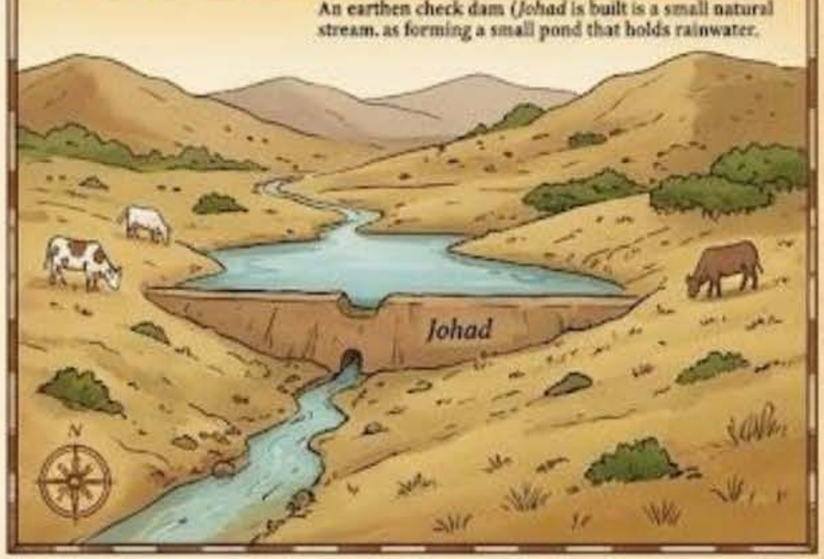
Gangetic Plains: *Baolis/Dighis*

A stepwell. stoen stepwell, balled (*Baoli*) built in no is flat agricultural strew (*Dighi*), built an ear open and/or open pond.



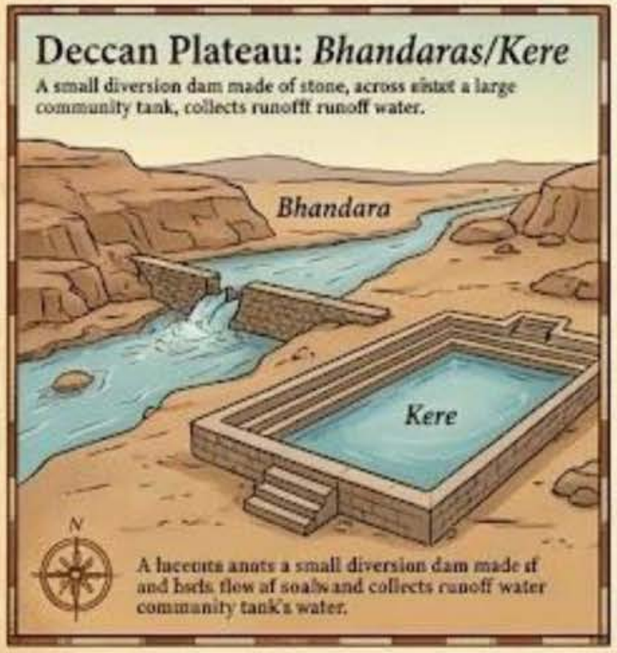
Central India: *Johads*

An earthen check dam (*Johad* is built is a small natural stream. as forming a small pond that holds rainwater.

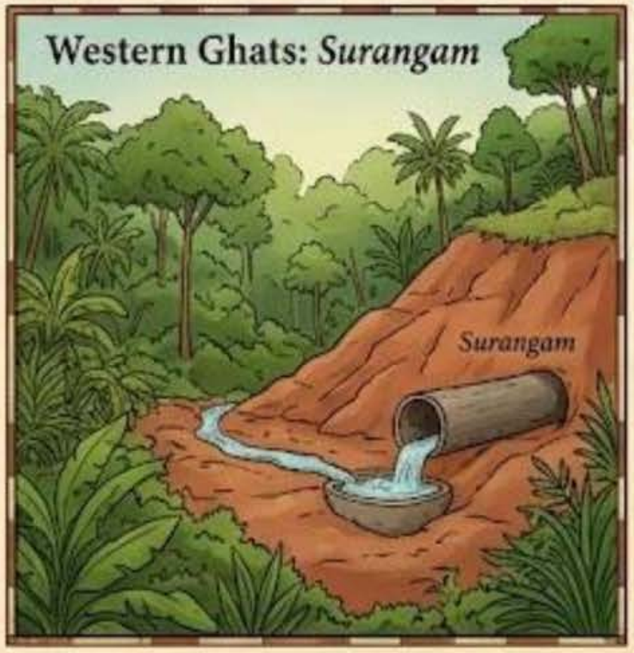


Deccan Plateau: *Bhandaras/Kere*

A small diversion dam made of stone, across sistat a large community tank, collects runofft runoff water.

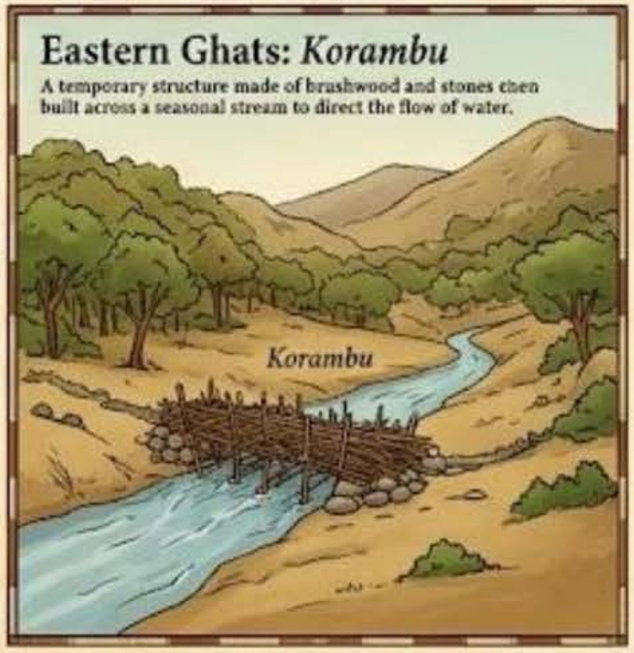


Western Ghats: *Surangam*



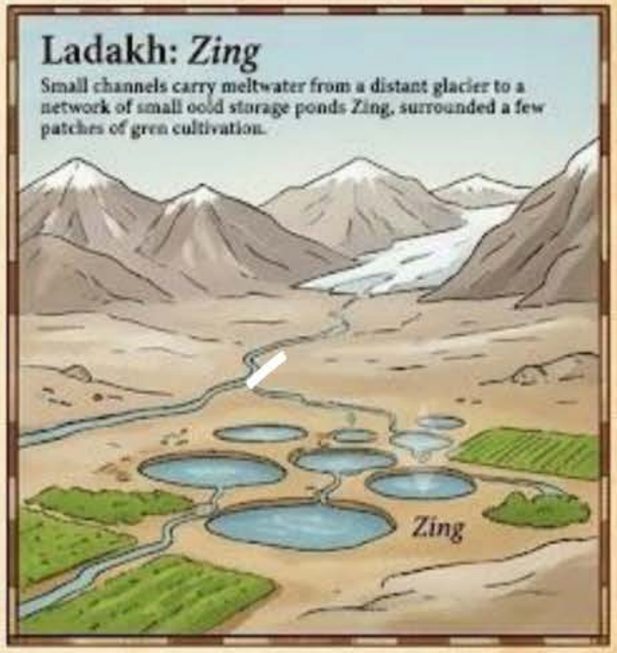
Eastern Ghats: *Korambu*

A temporary structure made of brushwood and stones chen built across a seasonal stream to direct the flow of water.



Ladakh: *Zing*

Small channels carry meltwater from a distant glacier to a network of small oold storage ponds Zing, surrounded a few patches of gren cultivation.



Need of Water Harvesting

- In India, we have acute shortage of water during the year.
- This is because we have not cared to conserve water or rationalised its use.
- The annual rainfall in India is 1,170mm (46 inches). This is higher compared to the world average of 800mm (32 inches).

Need of Water Harvesting

- Ironically, even Cherrapunji, which receives about 11,000 mm of rainfall annually, suffers from acute shortage of drinking water.

- This is because rainfall in India occurs in short spells of high intensity.

Need of Water Harvesting

- Due to such intensities coupled with short duration of heavy rain, most of the rain falling on the surface tends to flow away rapidly, leaving very little for the recharge of groundwater.

- This makes most parts of India experience lack of water even for domestic uses.

Which of the following is a ground water source?

(a) Pond
(b) Lake
(c) Spring
(d) River

Why is there a need to conserve water in India?

a) A lot of water is polluted
b) No water is found below the land in India
c) Water is in abundance in India
d) We receive lot of rain in summer

Assertion (A): Many states have recorded a sharp fall in the groundwater levels, ranging from 70 to 80per cent.
Reason (R): With the increase in population the demand for water has risen.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R does not explain A.
(c) A is true but R is false.
(d) A is false but R is true.

Which of the following is a cause for the decline in groundwater levels?

(a) Irregular and erratic rainfall
(b) Urbanisation
(c) Loss of vegetative cover
(d) All of the above.



Watershed management
= Conservation of ground & surface water.

Irrigation → सिंचाई

- Irrigation refers to the process of watering of agricultural plants through artificial means from wells, tanks, tubewells, canals, etc.
- Presently about 92per cent of water is used for agriculture, 2per cent for industries and 6per cent for drinking and domestic purposes.
- For farmers rainfall is still the most important source of water.
- About 55per cent of the net sown area in the country depends on rainfall to sustain crop production.

Irrigation

- In India, irrigation has been practised from ancient times.
- A number of artificial lakes and canals that dot the country are centuries old and some of them have served for more than a thousand years.

Irrigation

- The notable examples are the two big tanks in **Chingleput district of Tamil Nadu constructed in 8th and 9th centuries AD and the Grand Anicut built across the Cauveri river during the second century AD.**
- The irrigation canals built by various rulers from time to time, testify the importance of irrigation in the development of agriculture in India.



Irrigation

Why irrigation is required ?

1. **Uncertainty of Rainfall:** The rainfall in India is highly irregular in place and time, that is, normal rainfall is marked by wide fluctuations in different parts of the country. Sometimes the monsoon arrives early and at times it comes late.

Irrigation

2. **Uneven Distribution of Rainfall:** In India, the distribution of rainfall is quite uneven. Most parts of the country receive 80per cent of the annual rainfall from June to September from the South-west monsoon. Western Ghats and north eastern India receive more than 250 cm of annual rainfall, whereas the northern plains and the eastern Peninsula receive between 100 and 200cm of rainfall. The Kutch region of Gujarat, parts of Punjab and Haryana and western half of Rajasthan lie in the arid zones. Here the amount of rainfall is low and there is a constant scarcity of water. This inadequacy of rainfall has to be met by irrigation.

Irrigation

3. **Crop Requirements:** Different crops require different quantities of water for their growth.

(i) Most of the crops do not require water when they are maturing. However, grain crops require more water during the time ear heads are formed.

(ii) India has three cropping seasons-kharif, rabi and zaid. The kharif crops which occupy the largest area in the country are dependent on the monsoon rains. Where irrigation facilities are not developed, these crops are grown as rainfed crops.

Irrigation

- Rabi crops are grown on the soil moisture which is left after the harvest of kharif crops, or over the area where assured supply of irrigation is available.
- Zaid or summer crops are exclusively irrigated crops.
- Commercial crops like sugarcane need higher quantity of water as they require frequent watering which could be managed through efficient irrigation system.

Irrigation

4. **Nature of the Soil:** The water requirement of different crops depends on the nature of the soil. For example, clayey soils have higher moisture holding capacity and therefore, require less irrigation. On the other hand, the sandy loams have less moisture retentive capacity and need frequent watering, hence drip irrigation is suitable in dry regions.

Irrigation

5. **To Utilise River Water Effectively:** Many rivers in India are not perennial and carry insignificant flows during the rabi season. There is also a wide disparity in the water flow from year to year. The rivers of central and southern India have water for the four months of monsoon rains and remain largely dry during the eight months of the year. Therefore, to effectively utilise the water of these rivers means of irrigation have to be used.


INDIA
MAJOR RIVERS
N
TAJIKISTAN
AFGHANISTAN
PAKISTAN
CHINA (Tibet)
NEPAL
BHUTAN
BANGLADESH
MYANMAR (Burma)
LADAKH
JAMMU & KASHMIR
HIMACHAL PRADESH
CHANDIGARH
PUNJAB
UTTARAKHAND
HARYANA
DELHI
RAJASTHAN
UTTAR PRADESH
BIHAR
SIKKIM
ARUNACHAL PRADESH
ASSAM
NAGALAND
MEGHALAYA
MANIPUR
TRIPURA
MIZORAM
JHARKHAND
WEST BENGAL
MADHYA PRADESH
GUJARAT
MAHARASHTRA
CHHATTISGARH
ODISHA
TELANGANA
ANDHRA PRADESH
GOA
KARNATAKA
PUDUCHERRY
TAMIL NADU
KERALA
DADRA & NAGAR HAVELI AND DAMAN & DIU
Yanam (Puducherry)
Mahe (Puducherry)
Karaikal (Puducherry)
LAKSHADWEEP (INDIA)
ANDAMAN & NICOBAR ISLANDS (INDIA)
Arabian Sea
Bay of Bengal
MALDIVES
INDIAN OCEAN
SRI LANKA
Indus
Jhelum
Chenab
Ravi
Beas
Sutlej
Yarlung Tsangpo (Brahmaputra)
Yamuna
Gomati
Ganga
Ghaghara
Gandak
Kosi
Luni
Chambal
Banas
Betwa
Son
Sabarmati
Mahi
Narmada
Tapi
Parbati
Mahanadi
Brahmani
Subarnarekha
Hooghly
Padma
Meghna
Barak
Godavari
Bhima
Krishna
Indravati
Tungabhadra
Penner
Cauvery
Palar
Ponnaiyar
Kollidam
Irrawaddy (Irrawaddy)
International Boundary
State/UT Boundary
River
Map not to Scale
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Thank You